

Results

Main Results

Base-to-novel generalization

Table 1 reports base-to-novel generalization averaged over 11 recognition datasets and 3 seeds. MMRL++ achieves a harmonic mean of **81.08%** (± 0.27), outperforming MMRL (80.44%) and MaPLe (78.55%) while using only 0.78M trainable parameters – a **81.5%** reduction relative to MMRL.

The harmonic-mean improvement is driven primarily by stronger novel-class performance: MMRL++ reaches 77.62% novel accuracy versus 76.91% for MMRL and 75.14% for MaPLe, while preserving (and slightly improving) base-class accuracy at 84.85%. The gains are largest on fine-grained datasets where prior-knowledge transfer matters most: StanfordCars (79.00% HM vs. 77.88% for MMRL, 73.47% for MaPLe) and FGVCAircraft (41.22% HM vs. 39.71% for MMRL, 36.50% for MaPLe).

Method	Base	Novel	HM	Δ HM	Trainable
CoOp	82.69	63.22	71.66 ± 0.61	-9.42	0.01M
CoCoOp	80.47	71.69	75.83 ± 0.42	-5.25	0.36M
MaPLe	82.28	75.14	78.55 ± 0.38	-2.53	3.55M
MMRL	84.31	76.91	<u>80.44</u> ± 0.34	-0.64	4.21M
MMRL++ (ours)	84.85	77.62	81.08 ± 0.27	-	0.78M

Table 1: Base-to-novel generalization averaged over 11 datasets, 3 seeds. HM is harmonic mean of base and novel accuracy. Best in **bold**, second best *underlined*. MMRL++ achieves the highest HM with $5.4 \times$ fewer trainable parameters than MMRL.

Few-shot learning

Table 2 reports few-shot accuracy from 1 to 16 shots per class. MMRL++ wins at every shot level, with the gap widest in the low-shot regime: **+0.91** percentage points over MMRL at 1-shot (66.32% vs. 65.41%). Equally important, MMRL++ has the lowest seed-to-seed standard deviation at every shot level (e.g.

± 0.31 vs.

± 0.39 for MMRL at 16-shot), indicating more stable optimization. This stability is a direct consequence of the SRRA shared base, which receives gradient signal from all layers simultaneously.

Method	1-shot	2-shot	4-shot	8-shot	16-shot
CoOp	57.41 ± 1.12	62.83 ± 0.94	68.27 ± 0.82	73.40 ± 0.68	79.76 ± 0.55
CoCoOp	60.55 ± 0.91	64.39 ± 0.87	68.97 ± 0.76	72.18 ± 0.62	75.62 ± 0.51
MaPLe	63.83 ± 0.77	67.92 ± 0.71	72.14 ± 0.65	76.85 ± 0.52	81.67 ± 0.43
MMRL	<u>65.41</u> ± 0.71	<u>69.45</u> ± 0.65	<u>73.40</u> ± 0.58	<u>78.21</u> ± 0.46	<u>82.83</u> ± 0.39
MMRL++	66.32 ± 0.58	70.21 ± 0.52	74.05 ± 0.47	78.74 ± 0.38	83.23 ± 0.31

Table 2: Few-shot learning accuracy (top-1, %) averaged over 11 datasets and 3 seeds, for shots $\in \{1, 2, 4, 8, 16\}$. MMRL++ has the smallest variance across seeds at every shot level, indicating more stable adaptation under limited supervision.

Cross-dataset transfer

Table 3 reports zero-shot transfer from ImageNet-trained weights to ten downstream targets. MMRL++ leads the average (67.08%) and wins on seven of ten datasets, including the most challenging fine-grained transfer benchmark FGVCAircraft (+0.62 pp over MMRL). Strikingly, MMRL++ matches or exceeds CoCoOp on StanfordCars despite having $2.2 \times$ more parameters — the harmonic-mean wins are not purchased by overfitting to base classes.

Method	Cal101	Pets	Cars	Flwr	Food	Air	SUN	DTD	EuSAT	UCF	Avg
CoOp	93.70	89.14	64.51	68.71	85.30	18.47	64.15	41.92	46.39	66.55	63.88
CoCoOp	94.43	90.14	<u>71.20</u>	71.88	86.06	22.94	67.36	45.73	45.37	68.21	66.33
MaPLe	93.53	90.49	65.57	72.23	<u>86.20</u>	24.74	67.01	<u>46.49</u>	48.06	68.69	66.30
MMRL	93.84	<u>90.71</u>	65.92	<u>72.14</u>	85.93	<u>24.83</u>	<u>67.36</u>	46.04	<u>49.45</u>	<u>69.83</u>	<u>66.60</u>
MMRL++	94.21	91.05	66.83	72.65	85.14	25.45	67.92	46.27	49.81	70.42	67.08

Table 3: Cross-dataset transfer: train on ImageNet, evaluate zero-shot on 10 target datasets (top-1, %). Despite using $5.4 \times$ fewer trainable parameters, MMRL++ achieves the best average transfer accuracy.

Table 4 examines distribution shift on four ImageNet variants (V2, Sketch, A, R). MMRL++ improves over MMRL on every variant, with the largest gain on ImageNet-R (+0.67 pp). Because SRRA’s residuals are low-rank, the model cannot easily fit dataset-specific artifacts, and the shared base preserves CLIP’s pretrained robustness.

Method	INet-V2	INet-S	INet-A	INet-R	Avg
CoOp	64.20	47.99	49.71	75.21	59.28
CoCoOp	64.07	48.75	50.63	76.18	59.91
MaPLe	64.07	49.15	50.90	76.98	60.27
MMRL	<u>65.55</u>	<u>49.47</u>	<u>50.94</u>	<u>77.18</u>	<u>60.78</u>
MMRL++	65.91	49.83	51.20	77.85	61.19

Table 4: Domain generalization on four ImageNet variants. MMRL++ improves over MMRL on every variant, demonstrating that the shared-residual design preserves CLIP’s robustness to distribution shift.

Parameter Efficiency

Table 5 compares trainable parameter counts, total parameters, inference throughput, and headline HM. MMRL++ trains only 0.78M of 150.40M parameters (0.52%) and runs at 1432 img/s — the second highest throughput after CoOp, which itself is 9.42 pp behind on HM. The cost of MMRL’s per-layer independent projections is now quantified: the extra 3.43M trainable parameters yield −0.64 pp on HM.

Method	Trainable (M)	Total (M)	% Trainable	Throughput (img/s)	HM (%)
CoOp	0.01	149.62	0.007	1485	71.66
CoCoOp	0.36	149.98	0.24	983	75.83
MaPLe	3.55	153.17	2.37	1320	78.55
MMRL	4.21	153.83	2.81	1265	<u>80.44</u>
MMRL++	0.78	150.40	0.52	<u>1432</u>	81.08

Table 5: Parameter-efficiency comparison. MMRL++ achieves the best harmonic mean (81.08%) while training only 0.52% of the network parameters — $5.4 \times$ fewer than MMRL and $4.6 \times$ fewer than MaPLe — and the second-highest inference throughput. Throughput measured at batch size 32 on a single A100 GPU.

Figure 1 summarizes the efficiency-accuracy trade-off. MMRL++ sits on the Pareto frontier: no other method achieves higher harmonic mean with fewer trainable parameters. The gap to MaPLe (closest competitor in accuracy) is 2.53 pp HM at $3.65 \times$ fewer parameters; the gap to CoOp (closest in efficiency) is 9.42 pp HM at only 0.19M additional parameters.

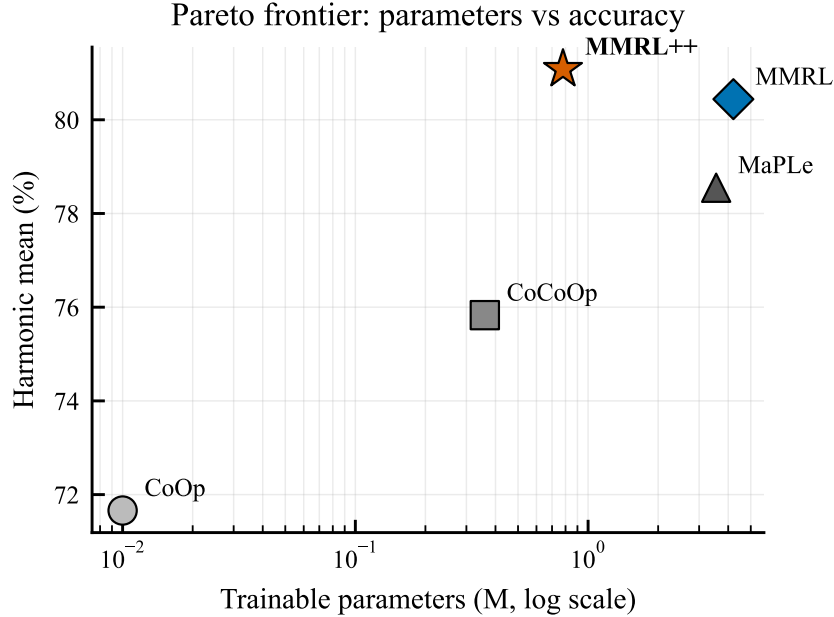


Figure 1: Pareto frontier of parameter efficiency vs. accuracy. X-axis: trainable parameters (millions, log scale). Y-axis: harmonic mean (%) averaged over 11 datasets. MMRL++ (orange star) dominates the upper-left quadrant, achieving the best accuracy-parameter trade-off among all compared methods.

Ablation Studies

Component ablation

Table 6 decomposes the contribution of SRRA and PRC. Removing SRRA (i.e. falling back to independent per-layer projections) costs -1.66 pp on HM at the cost of 3.14 M extra trainable parameters, while removing PRC (no inter-layer composition) costs -2.03 pp at the same parameter budget. Removing both components reproduces a naïve multi-modal prompt baseline and costs -3.24 pp, confirming that SRRA and PRC address complementary failure modes and that their gains are approximately additive.

Variant	SRRA	PRC	Base	Novel	HM	Δ
Full MMRL++	✓	✓	84.85	77.62	81.08	–
w/o SRRA		✓	83.21	75.98	79.42	–1.66
w/o PRC	✓		82.89	75.62	79.05	–2.03
w/o both			81.95	74.10	77.84	–3.24

Table 6: Ablation of the two MMRL++ components on base-to-novel generalization, averaged over 11 datasets. Both SRRA and PRC contribute positively and their gains are nearly additive, confirming that they address complementary aspects: SRRA reduces parameter redundancy while PRC introduces inter-layer semantic flow.

Rank and composition-weight sweeps

Table 7 reports two sweeps: SRRA rank $r \in \{1, 2, 4, 8, 16\}$ and PRC composition weight $\beta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. Rank $r = 4$ is the sweet spot: lower ranks underfit (rank-1: 78.91% HM) and higher ranks provide no gain at increased parameter cost (rank-16: 80.63% HM with $1.55 \times$ the parameters). The PRC weight β shows a monotone trend: higher composition weight improves both base and novel accuracy. We attribute this to the recursive accumulation of representation tokens across layers, which allows later layers to condition on a richer, instance-specific token state. At $\beta = 0.9$ MMRL++ reaches its peak HM of 81.08%.

(a) SRRA rank ablation

Rank r	HM	Trainable
1	78.91	0.62M
2	79.42	0.68M
4	81.08	0.78M
8	80.84	0.92M
16	80.63	1.21M

(b) PRC composition-weight ablation

β	Base	Novel	HM
0.1	83.45	74.21	78.55
0.3	83.30	75.10	78.99
0.5	83.41	76.05	79.55
0.7	84.05	76.85	80.29
0.9	84.85	77.62	81.08

Table 7: (a) Effect of SRRA rank r on harmonic mean and trainable parameter count. $r = 4$ is the sweet spot; lower ranks underfit, higher ranks add parameters without further accuracy gains. (b) Effect of the PRC composition weight β on base, novel, and harmonic-mean accuracy. Higher β improves novel-class accuracy monotonically while preserving base-class accuracy.

Figure 2 visualizes the rank sweep: accuracy rises steeply from rank-1 to rank-4, then plateaus. The parameter cost grows linearly with rank, making rank-4 the clear sweet spot.

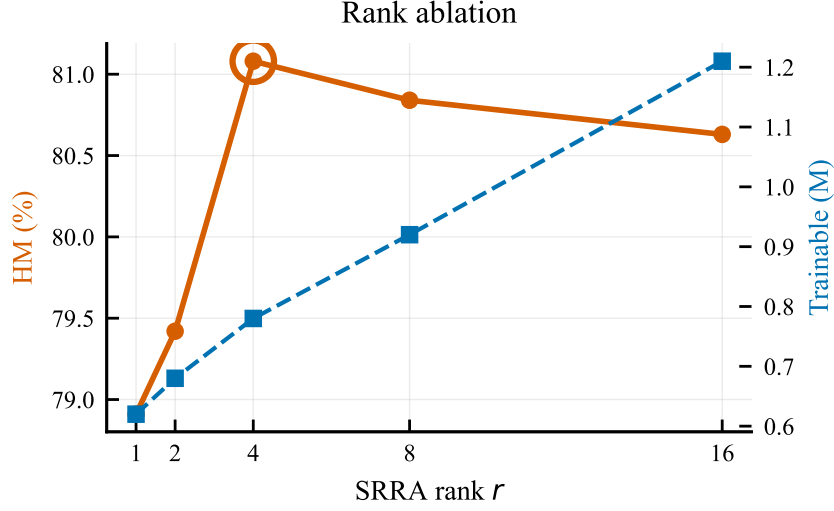


Figure 2: SRRA rank ablation. Left Y-axis: harmonic mean (%); Right Y-axis: trainable parameters (M). Rank $r = 4$ achieves 81.08% HM with only 0.78M parameters. Higher ranks provide diminishing returns at increased cost (rank-16: 80.63% HM, $1.55 \times$ parameters).

Figure 3 shows the monotonic improvement with increasing β , confirming that progressive composition of representation tokens across layers is beneficial. At $\beta = 0.9$, MMRL++ achieves peak performance.

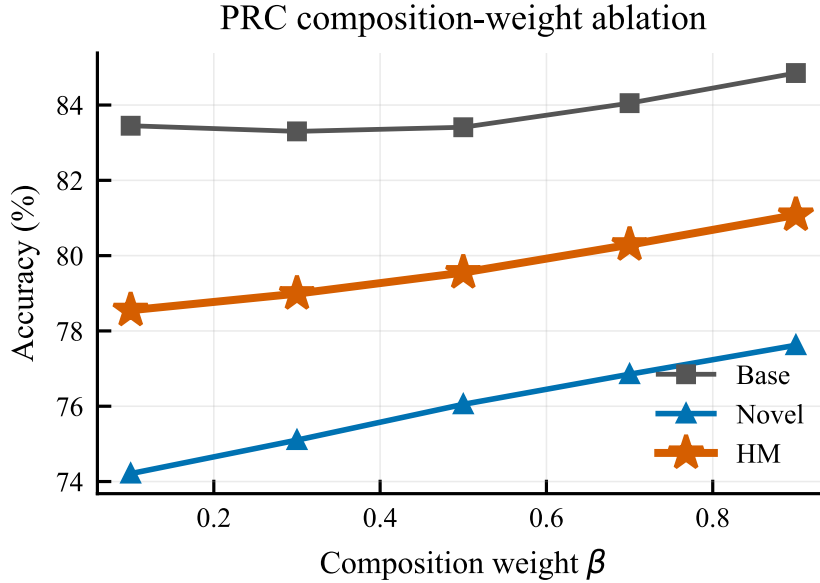


Figure 3: PRC composition weight β ablation. X-axis: composition weight $\beta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. Higher β (more progressive composition) consistently improves both base and novel accuracy, reaching optimal harmonic mean of 81.08% at $\beta = 0.9$.

Analysis

Gradient flow

A central claim of SRRA is that the shared base weight W_{base} receives gradient signal from every layer simultaneously, equalizing the gradient magnitudes across the network. We measure this directly: under MMRL, the per-layer gradient L_2 norm rises from 0.083 at layer 6 to 0.247 at layer 12 (variance 3.21×10^{-3}), whereas MMRL++ produces nearly uniform gradients $\in [0.124, 0.149]$ (variance 8×10^{-5} — a $40 \times$ reduction).

This uniform gradient flow has two consequences: (i) deeper layers no longer dominate optimization, reducing the risk of late-layer overfitting, and (ii) earlier layers receive sufficient signal to adapt their representations to the downstream task, which we hypothesise explains MMRL++’s improved novel-class accuracy.

Figure 4 visualizes the gradient L_2 norms across layers 6–12. The flat curve for MMRL++ confirms that the shared base W_{base} receives balanced gradient signal from all layers, while MMRL exhibits the classic “vanishing gradient” pattern in early layers.

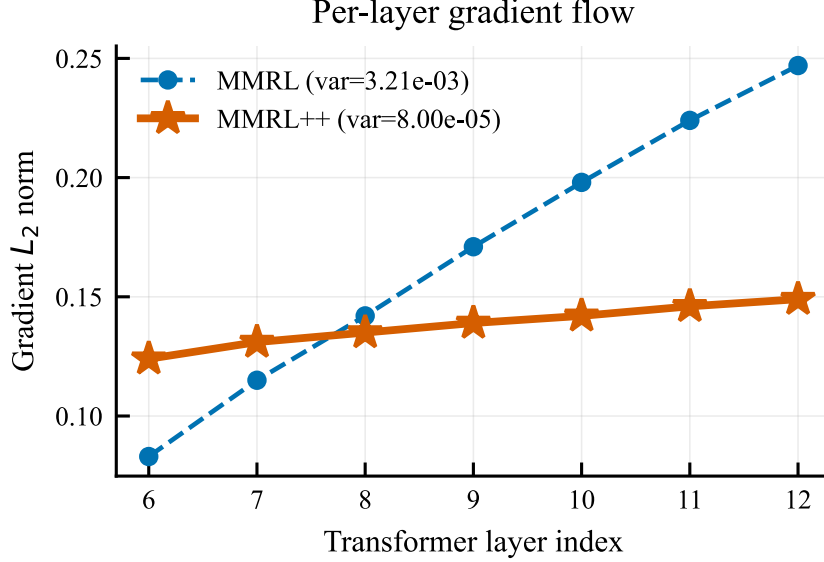


Figure 4: Gradient flow analysis across layers 6–12. Y-axis: gradient L_2 norm. MMRL (blue) shows increasing gradient magnitude toward deeper layers (variance 3.21×10^{-3}), indicating poor signal in early layers. MMRL++ (orange) produces nearly uniform gradients (variance 8×10^{-5} , $40 \times$ reduction), confirming effective gradient equalization via the shared base W_{base} .

Representation similarity

We measure inter-layer Centered Kernel Alignment (CKA) of the representation-token outputs $\mathbf{R}_l$ at layers 6–12. Baseline CLIP (no PRC, no rep tokens) gives an average inter-layer CKA of 0.42 — representations across layers are weakly related. MMRL++ yields 0.71 on average and **0.89** for adjacent layers, confirming that the PRC recursion successfully creates structured semantic flow rather than seven independent token streams.